

Customer Purchasing Behavior & Product Recommendation Analysis

Snapdeal | Python-Based Customer Insights Project

PROJECT PURPOSE

Analyze customer purchasing behavior, shopping satisfaction, product category preferences, recommendation engagement and other behavioral segments to identify actionable insights for personalization, retention and customer experience improvement.

1. Executive Summary

This project analyzes an 800-response Snapdeal customer survey dataset to understand purchasing behavior, browsing patterns, shopping satisfaction, recommendation engagement, review perceptions, and customer segments. The analysis was completed in Python using data cleaning, descriptive statistics, visualization, rule-based segmentation, and K-Means clustering.

The analysis indicates that customer behavior is relatively distributed across demographic groups, while purchase frequency and satisfaction provide a clear basis for distinguishing customer segments. Clothing and Fashion is the most represented purchase category after splitting multi-category responses, and price-related factors are prominent among cart abandonment reasons.

KEY TAKEAWAY

Purchase frequency and shopping satisfaction emerged as the strongest dimensions for behavioral segmentation. A four-cluster K-Means solution provided the most interpretable structure, separating customers into high/low frequency and high/low satisfaction patterns.

Top Findings

- The cleaned analysis dataset contains 800 customer responses. No duplicate rows were found.
- Clothing and Fashion represented 22.97% of category participation after multi-category purchase responses were split and exploded.
- Purchase frequency was broadly distributed; 20.75% reported purchasing multiple times a week, while 20.50% purchased once a month.
- The most common cart-abandonment factor was finding a better price elsewhere (27.38%), followed by other factors (24.50%) and high shipping costs (24.25%).
- The correlation between recommendation helpfulness and shopping satisfaction was only 0.04, indicating a negligible linear relationship in this dataset.
- Review helpfulness and review reliability each showed only a 0.02 correlation with shopping satisfaction in the analysis.
- Personalized recommendation frequency declined from satisfaction level 1 to level 5, with counts of 173, 166, 163, 155, and 143 respectively.
- K-Means identified four interpretable behavioral clusters, primarily differentiated by Shopping_Satisfaction and Purchase_Frequency_Level.

2. Project Overview

Business Context

Snapdeal leadership wants to understand how customers shop, what categories they purchase, how they engage with recommendations and reviews, and how customer behavior differs across segments. The objective is to use these insights to improve personalized shopping experiences and support marketing, retention, and inventory decisions.

Objectives

1. Clean and prepare the customer survey data for reliable analysis.
2. Describe customer demographics, purchase frequency, category preferences, browsing behavior, cart abandonment, and satisfaction.
3. Create actionable customer segments and profile their demographic and behavioral differences.
4. Evaluate relationships between recommendation helpfulness, shopping satisfaction, review perceptions, and personalized recommendation engagement.
5. Translate findings into practical recommendations and a clear reporting structure.

3. Data Preparation & Quality

The source file initially contained 800 records and 24 columns. Data-quality checks found no duplicate rows. Column names were standardized by stripping extra whitespace, and an unwanted duplicate/index column was removed during preparation.

Data Quality Area	Action Taken	Outcome
Duplicate rows	Checked using <code>duplicated().sum()</code>	No duplicate rows found
Column formatting	Stripped extra spaces from column names	Cleaner, consistent names
Missing values	Profiled missing values by column	149 missing values identified in <code>Product_Search_Method</code>
Product search method	Added 'Unknown' for missing responses	Missing category retained explicitly
Duplicate column	Removed duplicated <code>Personalized_Recommendation_Frequency</code> field	Eliminated redundant field
Numeric fields	Verified integer/numeric types for ratings and age	Ready for analysis
Purchase categories	Split semicolon-separated multi-category responses and exploded rows	Enabled individual category popularity analysis

4. Descriptive Customer Behavior

Customer Demographics

Gender	Share of Responses
Male	26.6%
Female	25.1%
Prefer not to say	24.2%
Others	24.0%

The gender distribution is relatively balanced, with Male respondents representing the largest group at 26.6%. Age ranged from 3 to 67 years, with a mean of 36.05 and a median of 37.

Purchase Frequency

Purchase Frequency	Share
Multiple times a week	20.75%
Once a month	20.50%

Once a week	20.13%
Few times a month	19.62%
Less than once a month	19.00%

Purchase Categories

Category	Participation
Clothing and Fashion	22.97%
Beauty and Personal Care	19.98%
Home and Kitchen	19.55%
Others	19.11%
Groceries and Gourmet Food	18.39%

Because customers could report multiple purchase categories in one response, the category field was split into individual categories before calculating popularity. Clothing and Fashion had the highest category participation at 22.97%, followed by Beauty and Personal Care at 19.98%.

Browsing & Cart Abandonment

Finding	Share
Multiple times a day browsing	25.50%
Few times a week browsing	25.00%
Few times a month browsing	24.88%
Rarely browsing	24.38%
Better price elsewhere — cart abandonment	27.38%
Other cart-abandonment factors	24.50%
High shipping costs — cart abandonment	24.25%

Browsing frequency is relatively evenly distributed, while price sensitivity stands out as the leading stated cart-abandonment factor. This supports stronger price-value communication and targeted promotional strategies.

Satisfaction & Rating Measures

Metric	Mean	Median
Shopping Satisfaction	2.99	3.00
Recommendation Helpfulness (encoded)	1.96	2.00
Rating Accuracy	2.96	3.00

Shopping satisfaction was centered around the midpoint, with a mean of 2.99 and median of 3. The analysis therefore indicates a mixed rather than overwhelmingly positive satisfaction profile.

5. Customer Segmentation & Profiling

Two segmentation approaches were explored. First, business-rule segments were created using purchase frequency and shopping satisfaction. Second, K-Means clustering was applied to the two strongest behavioral dimensions identified in the analysis: Shopping_Satisfaction and Purchase_Frequency_Level.

Rule-Based Segments

Segment	Definition Used	Share
Frequent Shopper	Purchase frequency level ≥ 4 and satisfaction ≥ 4	16.00%
Occasional Shopper	Purchase frequency level ≥ 2 and satisfaction ≥ 2 , outside Frequent Shopper	80.12%
At Risk	Purchase frequency level < 2 and satisfaction < 2	3.88%

The rule-based approach showed that the majority of respondents fall into the Occasional Shopper category, creating a large opportunity to increase purchase frequency through targeted engagement.

K-Means Behavioral Clusters

Cluster	Size	Avg. Satisfaction	Avg. Purchase Frequency	Business Profile
Cluster 0	251	4.13	3.89	Loyal / Frequent Buyers
Cluster 1	189	1.51	2.08	At-Risk / Disengaged
Cluster 2	194	3.95	1.48	Satisfied Occasional Shoppers
Cluster 3	166	1.83	4.63	Frequent but Dissatisfied

SEGMENTATION INSIGHT

The four clusters create a clear behavioral matrix: high frequency/high satisfaction (Cluster 0), low frequency/high satisfaction (Cluster 2), high frequency/low satisfaction (Cluster 3), and lower frequency/lower satisfaction (Cluster 1). This is useful because frequent-but-dissatisfied customers are distinct from low-frequency at-risk customers and require a different intervention.

Segment-Specific Recommendations

Segment	Recommended Action	Business Objective
Loyal / Frequent Buyers (0)	Retain with loyalty benefits, personalized cross-sell and exclusive offers	Increase lifetime value
At-Risk / Disengaged (1)	Re-engagement campaigns, targeted incentives and friction reduction	Recover customers / reduce churn
Satisfied Occasional (2)	Personalized offers, complementary products and repeat-purchase incentives	Increase purchase frequency
Frequent but Dissatisfied (3)	Prioritize service recovery and identify dissatisfaction drivers before pushing promotions	Protect high-value revenue

6. Recommendation & Review Insights

Recommendation Helpfulness vs Shopping Satisfaction

Recommendation helpfulness and shopping satisfaction showed a correlation of 0.04. This is a negligible relationship, suggesting that overall satisfaction in this dataset does not depend primarily on recommendation helpfulness. However, the group comparison showed that customers who reported recommendations as 'No' had lower average satisfaction than customers reporting 'Sometimes' or 'Yes'.

Recommendation Helpfulness	Average Shopping Satisfaction
No	2.89
Sometimes	3.08
Yes	3.02

Review Reliability & Helpfulness

Review helpfulness and review reliability each showed only a 0.02 correlation with shopping satisfaction. Average satisfaction was also broadly similar across review-helpfulness classes and review-reliability levels, although the highest reliability level had a slightly higher average satisfaction.

Personalized Recommendation Engagement

Frequency Level	Count	Share
1	173	21.62%
2	166	20.75%
3	163	20.37%
4	155	19.38%
5	143	17.88%

Personalized recommendation frequency declined steadily from level 1 to level 5. The analysis also found that levels 1 and 5 had lower average satisfaction (2.84 and 2.92 respectively) than the middle levels. The relationship between recommendation frequency and satisfaction was negligible (0.017).

Recommendation Engagement by K-Means Cluster

Cluster	Avg. Personalized Recommendation Frequency	Avg. Satisfaction
0	2.89	4.13
1	2.92	1.51
2	3.00	3.95
3	2.84	1.83

Cluster 2 (Satisfied Occasional Shoppers) had the highest average personalized recommendation frequency, while Clusters 0 and 3 — both high-purchase-frequency groups — had lower average recommendation frequency. This creates an opportunity to improve personalization for high-frequency customers, especially Cluster 3.

7. Visualization & Reporting Highlights

The notebook contains visualizations covering purchase category participation, browsing frequency, satisfaction levels, cart-abandonment factors, customer segmentation, recommendation engagement, and correlation analysis. These visuals support the numerical findings and make segment differences easier to communicate to business stakeholders.

Visualization	Primary Business Question
Purchase Category Distribution	Which categories have the highest customer participation?
Browsing Frequency Distribution	How frequently do customers browse?
Satisfaction Distribution	Where is the customer satisfaction profile concentrated?
Recommendation–Satisfaction Heatmap	Is recommendation helpfulness strongly associated with satisfaction?
Customer Segment Distribution	How large is each behavioral segment?
Cluster Profile Comparison	How do the four behavioral groups differ?
Personalized Recommendation Frequency	How actively do customers engage with personalization?

8. Strategic Recommendations

1. Increase the frequency and quality of personalized recommendations for high-purchase-frequency customers, particularly Clusters 0 and 3, using purchase history, browsing behavior, and category preferences.
2. Prioritize recommendation relevance over simply increasing message volume. The 0.04 correlation between recommendation helpfulness and satisfaction indicates that recommendation quality alone does not explain overall satisfaction.
3. Use segment-specific recommendation strategies. Satisfied occasional shoppers can receive cross-sell and repeat-purchase nudges, while frequent but dissatisfied customers should receive service-recovery interventions before additional promotional pressure.
4. Strengthen review trust through verified reviews, clear product information, and rating-quality controls. The weak 0.02 relationships indicate that review perceptions alone are not major drivers of satisfaction in this dataset.
5. Address price sensitivity and cart abandonment by improving price-value communication, targeted discounts, competitive pricing where appropriate, and shipping-cost transparency.
6. Use Clothing and Fashion as a priority category for inventory, while exploring complementary relationships with Beauty and Personal Care and Home and Kitchen.

9. Conclusion

The Snapdeal customer insights analysis demonstrates how customer survey data can be transformed into actionable behavioral intelligence. The analysis identified major purchase categories, browsing and cart-abandonment patterns, satisfaction characteristics, and four distinct behavioral customer clusters.

The strongest segmentation signal came from the combination of purchase frequency and shopping satisfaction. This provides Snapdeal with a practical framework for differentiating loyal customers, satisfied but occasional shoppers, frequent but dissatisfied customers, and lower-engagement customers.

The recommendation analysis further shows that simply increasing recommendation frequency is unlikely to solve satisfaction challenges. Instead, Snapdeal should focus on relevance, segment-specific personalization, service recovery for dissatisfied high-frequency customers, and targeted strategies that convert satisfied occasional shoppers into more frequent customers.

Ending Note

The highest-value opportunity is not treating all customers the same. Snapdeal can improve customer value by matching the intervention to the customer's behavioral profile — retain the loyal, grow the satisfied occasional, recover the frequent-but-dissatisfied, and selectively re-engage the at-risk customer segments.

VIDEO LINK – <https://drive.google.com/drive/folders/1XVvOLsJ616mMi4SmwFzeEUE47-tjNgpl?usp=sharing>